Bivariate Regression II

October 31, 2022

As you come in...

```
Call:
lm(formula = Dem.Pres.candFT ~ FemFT, data = anes)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.33745    1.16472   4.583 4.75e-06 ***
FemFT        0.68409    0.01885  36.298  < 2e-16 ***
```

- Interpret the $\hat{\beta}$. The DV is Hillary Clinton Feeling Thermometer, and the IV is the Feminists Feeling Thermometer from ANES
- Is $\hat{\beta}$ significant?
- Predict $\hat{Y}$ when X is 10
- Predict $\hat{Y}$ when X is 0

Calculating Regression Parameters

$$Y = \alpha + \hat{\beta} X \tag{1}$$

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$$\hat{\beta} = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sum(x_i - \bar{x})^2} \tag{3}$$

You must know $\hat{\beta}$ before you can calculate $\hat{\alpha}$

Remember this?

$$cov_{XY} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

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So if Covariance increases, β increases, and if X is less spread out, β decreases (slope becomes flatter)

Y	**X**
Family Income (in thousands)	Years of Education
30	10
40	12
40	15
50	12
50	16
65	16
65	21
80	19
100	16
100	20
$\bar{Y} = 62$	$\bar{X} = 15.7$

$$
\begin{aligned}
\sum\left[(x-\bar{x})(y-\bar{y})\right] = \quad & (10-15.7)(30-62) \\
& +(12-15.7)(40-62) \\
& +(15-15.7)(40-62) \\
& +(12-15.7)(50-62) \\
& +(16-15.7)(50-62) \\
& +(16-15.7)(65-62) \\
& +(21-15.7)(65-62) \\
& +(19-15.7)(80-62) \\
& +(16-15.7)(100-62) \\
& +(20-15.7)(100-62) \\
& = 571
\end{aligned}
$$

$$
\begin{aligned}
\sum(x-\bar{x})^2 &= (10-15.7)^2 \\
&+(12-15.7)^2 \\
&+(15-15.7)^2 \\
&+(12-15.7)^2 \\
&+(16-15.7)^2 \\
&+(16-15.7)^2 \\
&+(21-15.7)^2 \\
&+(19-15.7)^2 \\
&+(16-15.7)^2 \\
&+(20-15.7)^2 \\
&= 118.1
\end{aligned}
$$

$$\hat{\beta} = \frac{\sum\left[(x-\bar{x})(y-\bar{y})\right]}{\sum(x-\bar{x})^2} = \frac{571}{118.1} = 4.83$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 62 - 4.83 \times 15.7 = -13.83$$

What is our regression equation?

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What is our regression equation?

$$Y = -13.83 + 4.83X$$

$$\text{Income} = -13.83 + 4.83(\text{Education})$$

How do we know if X predicts Y?

We can do a hypothesis test!

- Null... $H_0 : \beta = 0$
- Alternative... $H_A : \beta \neq 0$

$$t = \frac{\hat{\beta} - \beta_{null}}{SE(\hat{\beta})} = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

If we reject the null (using p-values or confidence intervals or the critical value approach), we conclude there is a statistically significant relationship between X and Y.

Goodness of Fit: How well is $\boldsymbol{X}$ explaining $\boldsymbol{Y}$?

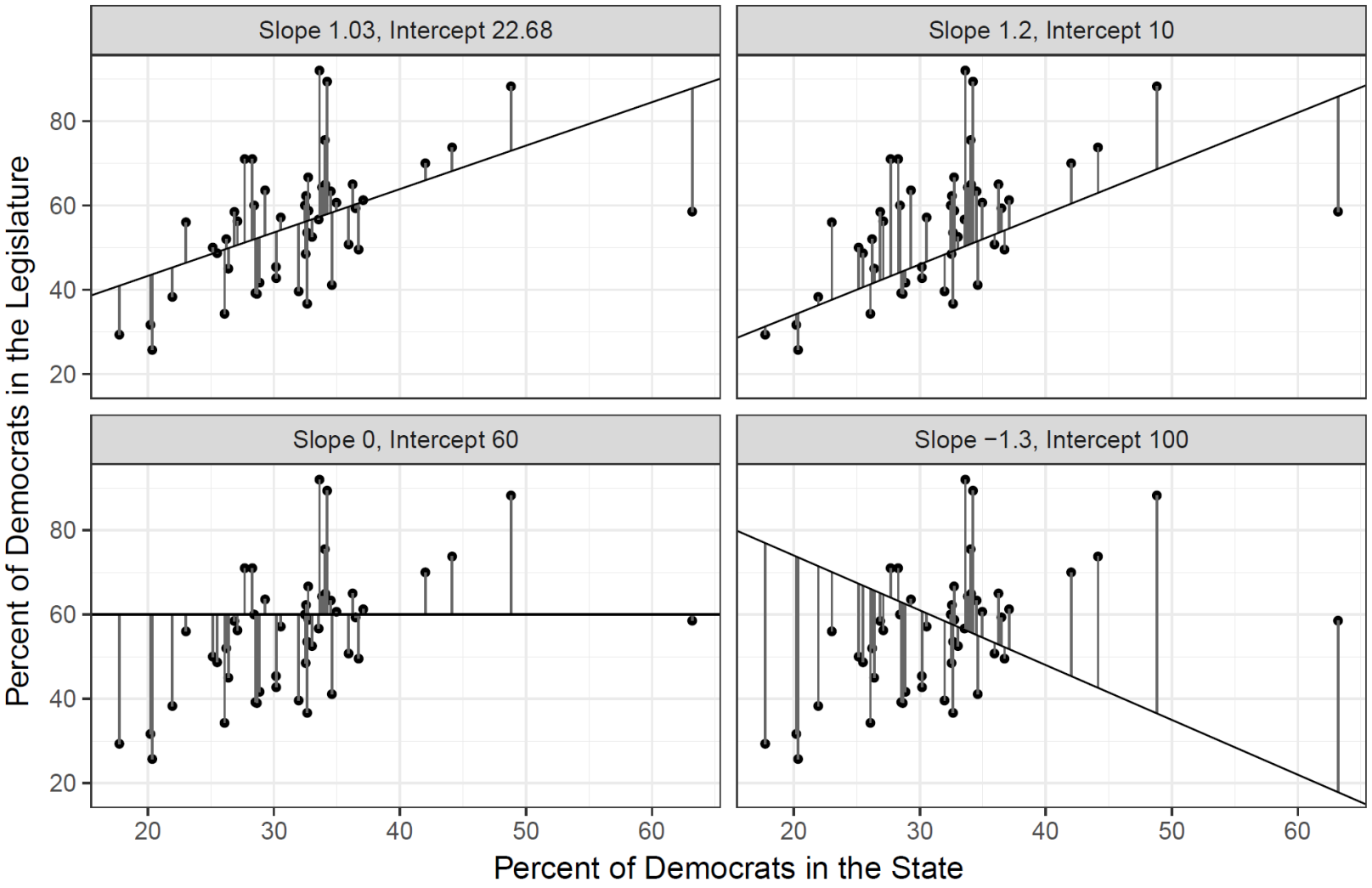
Measure I (Probably Won't Use): RMSE

Root Mean-Squared Error: standard error of the regression model

- Measures the average accuracy of the model
- Essentially the average distance between the data points and the regression line

$$RMSE = \sqrt{\frac{\sum \hat{\epsilon}^2}{n}} = \sqrt{\frac{\sum (Y_i - \hat{Y}_i)^2}{n}} \tag{4}$$

- Number measured in terms of DV (e.g., if rmse is 4, then on average model is off by 4 units of DV)
- Not robust to outliers though


Slope 1.03, Intercept 22.68
Slope 1.2, Intercept 10
Slope 0, Intercept 60
Slope −1.3, Intercept 100
Percent of Democrats in the Legislature
Percent of Democrats in the State
20
40
60
80
20
30
40
50
60

Measure II: Residual Standard Deviation

Standard deviation of the residuals, so measure of dispersion of the errors

$$\hat{\sigma} = \frac{\sum(Y_i - \hat{Y})^2}{n-2} \tag{5}$$

Interpret like a standard deviation (95% of the data fall within 2 residual s.d. of regression line)

- Generally, <u>smaller</u> is better

Measure III: R^2 Venn Diagram

What if I want to just know how much of Y I'm explaining with X?

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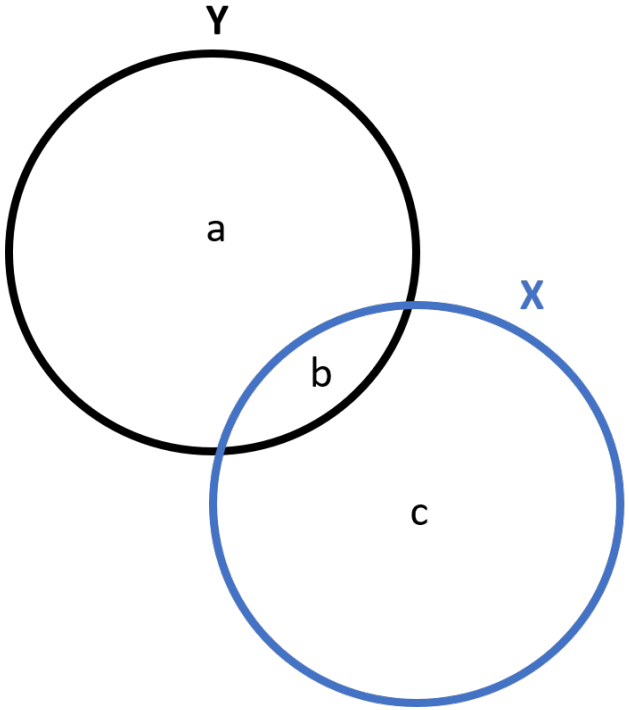

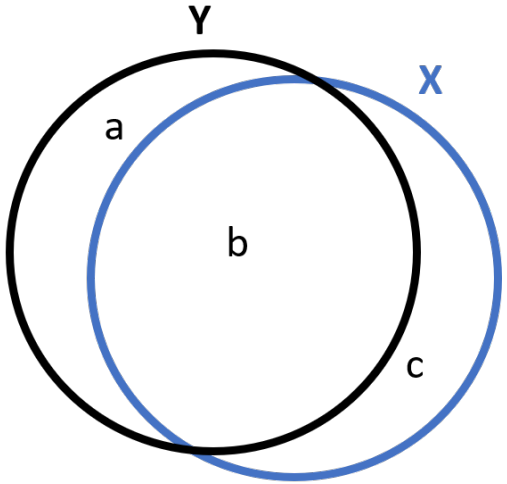

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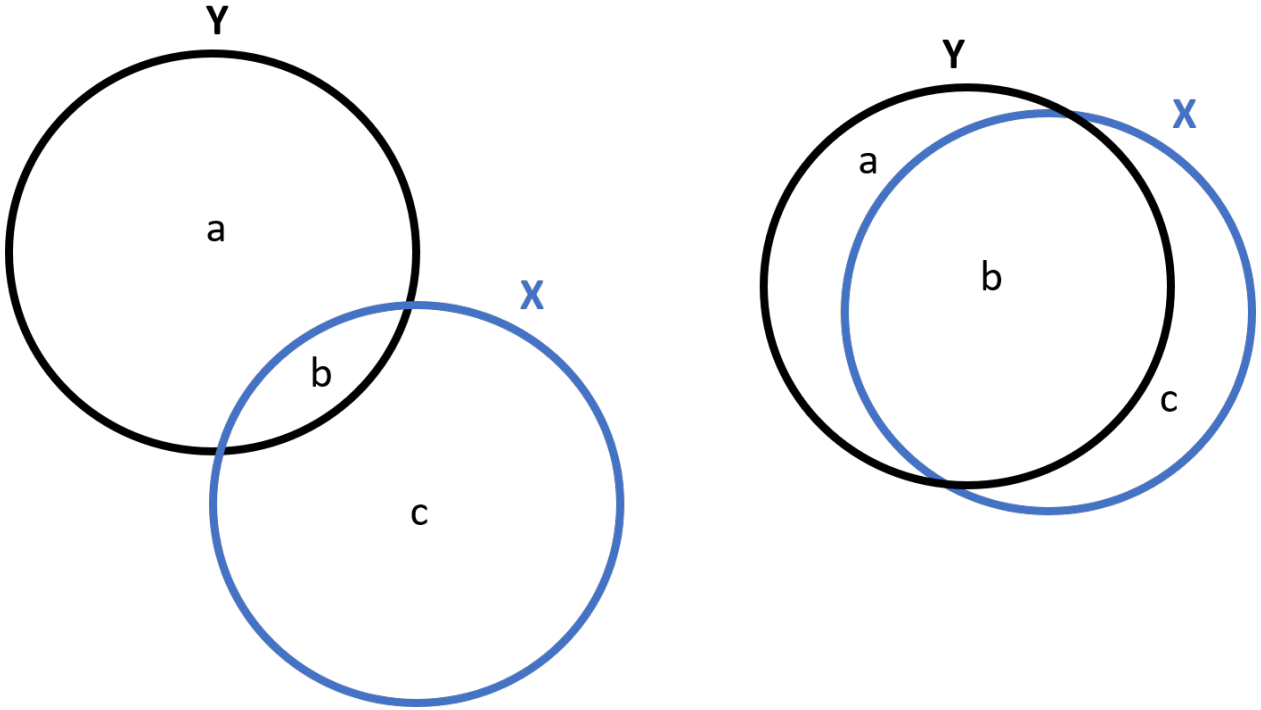

$$R^2 = \frac{b}{Y} = \frac{\text{Model Sum of Squares}}{\text{Total Sum of Squares (TSS)}} \tag{6}$$

Measure III: R^2

$$R^2 = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2} \tag{7}$$

What proportion of the variation in y is explained by your model?

- Between 0 and 1
- Generally, <u>bigger</u> is better

Example: $R^2 = 0.27$ means "the model explains 27% of the variation in Y."

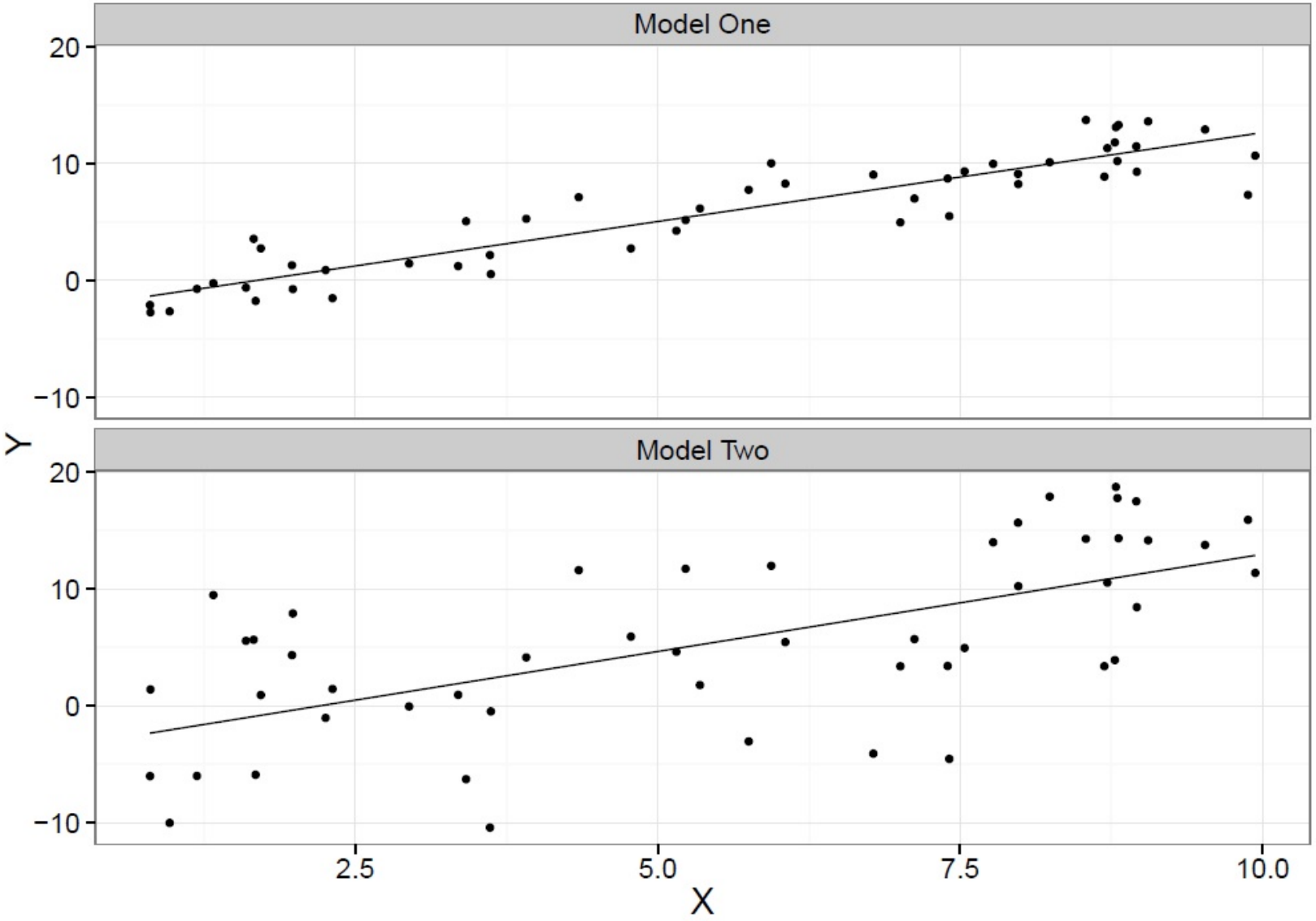
Model One
Model Two
20
10
0
−10
Y
2.5
5.0
7.5
10.0
X

Table: Model Fit Comparison

	Model 1	Model 2
(Intercept)	-2.56*	-3.66*
	(0.58)	(1.79)
X	1.52*	1.66*
	(0.09)	(0.29)
N	50	50
R^2	0.85	0.41
Resid. sd	1.94	6.00

Standard errors in parentheses

* indicates significance at $p < 0.05$

	Dependent variable:		
	% Dems in Leg.		
	(1)	(2)	(3)
% Democrat	1.032*		
	(0.252)		
% Republican		−1.518*	
		(0.237)	
% Nonwhite			0.307*
			(0.131)
Constant	22.684*	103.167*	48.381*
	(8.215)	(7.600)	(3.652)
Observations	49	49	49
R^2	0.263	0.467	0.105
Residual Std. Error	12.944	11.015	14.269

Major Assumptions

1. **Y** and **X** are linearly related
2. Errors (residuals, u_i, ϵ_i, etc.) are $\sim \mathcal{N}(0, \sigma^2)$
   - Errors have a mean of 0, so not biased
   - Variance is σ^2 and homoscedastic
   - No autocorrelation
3. No measurement error in **X**
4. No causal variable left out, and no non-causal variables are included

Fitting a Regression in R (Generally)

regression$< -$lm(formula $=$ **y-variable** $\sim$ **x-variable**, data $=$ **datasetname**)

summary(regression)